CSI 4106. Introduction to Articial Intelligence

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Course info

	Day	Time	Location
Lecture 1	Monday	13:00-14:20	CRX C240
Lecture 2	Wednesday	11:30-12:50	CRX C240
Office hours	Monday	15:00-16:20	STE 5106

Description (official)

The roots and scope of Artificial Intelligence. Knowledge and knowledge representation. Search, informed search, adversarial search. Deduction and reasoning. Uncertainty in Artificial Intelligence. Introduction to Natural Language Processing. Elements of planning. Basics of Machine Learning.

Approach

Machine learning first. Unlike other artificial intelligence courses, this one is structured so that deep learning is presented as early as possible. This serves two purposes. First, this is because deep learning is such a dominant technology now that it would be difficult to grab students attention without studying this subject. Second, deep learning serves as a framework through which we can introduce and define key topics in artificial intelligence. In some cases, we will see how deep learning has displace other technologies, or replace parts of it. Whereas in other cases, this will be an opportunity to discuss the limitations of deep learning and see how previous technologies had been developed for solving specific problems. Finally, it is important to note that learning itself represents one of the earliest and most extensively understood milestones in the evolution of intelligence.

Learning outcomes

Upon completion of the course, you will be able to:

- **Explain** the fundamental concepts of Artificial Intelligence (AI)
- **Apply** problem-solving strategies using AI techniques
- **Critically analyze** and **compare** different AI approaches
- **Demonstrate** independent learning and exploration

Outline

Given the widespread influence of deep learning in current Artificial Intelligence advancements, my aim is to incorporate it into the course curriculum at an early stage. Establishing this groundwork will facilitate our understanding of its significance, particularly as we delve into subjects such as Monte Carlo Tree Search (MCTS) later in the course. Below is a preliminary and ambitious course outline.

- Defining AI
- Introduction to machine learning
- Linear regression
- Training
- Logistic regression
- Evaluation
- Hyperparameter tuning
- Machine learning engineering
- Introduction to artificial neural networks
- Backpropagation
- Softmax, cross-entropy, regularization
- Convolutional neural networks
- Introduction to search

- Informed search
- Local search
- Population-based metaheuristics
- Adversarial search
- Monte Carlo Tree Search

Grading

The final course grade will be calculated as follows:

Category	Percentage
Assignments	40% (4 x 10%)
Quiz	20%
Final examination	40%

Consult the schedule for the dates.

In order to pass the course, students must obtain a combined average of at least 50% on the quizzes and the final exam. Specifically, $(Q1 + Q2 + F)/60 \geq 0.50$, where $Q1$, $Q2$, and F denote the first and second quizzes, and the final examination scores, respectively.

Assignments, quizzes and the final examination constitute mandatory evaluation components. To be eligible to sit for the final examination, students must complete at least one of the two quizzes and at least two of the four assignments. Failure to satisfy these mandatory evaluation requirements will result in an automatic `EIN` final grade.

If you are unable to complete one quiz due to valid and documented reasons, such as illness with a doctor's note, the weight of that quiz will be transferred to the other quiz. Requests are required to be submitted within three (3) business days following the examination, in accordance with the academic regulations.

To accommodate circumstances beyond a student's control, such as illness or submission errors, the assignment component of the final grade is calculated from the student's three highest scores among the four assignments.

Bonus marks on assignments are intended to compensate exclusively for lower scores on other assignments, reflecting the fact that assignments and examinations test different competencies. Moreover, in today's AI-driven world, the significance of in-person evaluations is more pronounced than ever.

For additional information, particularly regarding queries about grade adjustments, please refer to the FAQ section.

Material and resources

- Lecture notes (slides) and complementary resources are posted on the course Web site: turcotte.xyz/teaching/csi-4106/

Monographs

This course does not require any mandatory textbook. However, those who wish to consult a general reference work may refer to the following:

- Russell, S., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.

In most cases, the course notes provided will be sufficient to follow the course.

Academic integrity

Academic fraud is an act by a student that may result in a false evaluation (including papers, tests, examinations, etc.). It is not tolerated by the University. Any person found guilty of academic fraud will be subject to sanctions.

Here are some examples of academic fraud:

- Plagiarism or cheating of any kind;
- Present research data that has been falsified;
- Submit a work for which you are not the author, in whole or part;
- Submit the same piece of work for more than one course without the written consent of the professors concerned.
- Please consult this webpage: it contains regulations and tools to help you avoid plagiarism.

An individual who commits or attempts to commit academic fraud, or who is an accomplice, will be penalized. Here are some examples of possible sanctions:

- Receive an "F" for the work or in the course in question;
- Imposition of additional requirements (from 3 to 30 credits) to the program of study;
- Suspension or expulsion from the Faculty.
- You can refer to the regulations on this web page

Student services

Academic writing help

At the Academic Writing Help Centre you will learn how to identify, correct and ultimately avoid errors in your writing and become an autonomous writer. In working with our Writing Advisors, you will be able to acquire the abilities, strategies and writing tools that will enable you to:

- **Master** the written language of your choice
- **Expand** your critical thinking abilities
- **Develop** your argumentation skills
- **Learn** what the expectations are for academic writing

Further information is available here:

- www.uottawa.ca/study/academic-support/academic-writing-help

Career services

Career Services offers various services and a career development program to enable you to recognize and enhance the employability skills you need in today's world of work.

- www.uottawa.ca/current-students/career-experiential-learning/career-development

Counselling service

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- www.uottawa.ca/campus-life/health-wellness/counselling-therap

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- www.uottawa.ca/study/academic-support/accommodation-services-available

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